

● HARD

● CRAFT AND STRUCTURE

Cross-Text Connections

10 questions · ~15 min

02

Text 1

When the 50-second-long film *Arrival of the Train*—which depicts what its title says, a train pulling into a station—was first shown publicly in 1896, spectators, naïve to the new medium of film and seeing a train appearing to come directly at them, leaped from their seats and fled the room. This moment marks a major cultural shift: a new way of representing and seeing the world had arrived with that train, and nothing would ever be the same.

Text 2

The fact that there is no contemporary evidence that the first audience of *Arrival of the Train* was alarmed has not stopped the story from becoming canonical, even among film historians. But that phenomenon itself is highly revealing. Our belief that the coming of film was transformative is so strong that we invented and keep retelling a founding myth that divides cultural history into a (naïve) “before” and (sophisticated) “after.”

Based on the texts, the author of Text 2 would most likely agree with which statement about the description of the first showing of *Arrival of the Train* in Text 1?

- A. It is more motivated by a perception of the significance of the invention of film than by facts.
- B. It expresses a view about the transformative nature of film that film historians tend to regard as naïve.
- C. It is not based on evidence and reflects film historians’ belief that they are more sophisticated than today’s audiences.
- D. It reflects a misconception that is widely believed because it captures early film audiences’ sense that the new medium was alarming.

Text 1

In 1916, H. Dugdale Sykes disputed claims that *The Two Noble Kinsmen* was coauthored by William Shakespeare and John Fletcher. Sykes felt Fletcher's contributions to the play were obvious—Fletcher had a distinct style in his other plays, so much so that lines with that style were considered sufficient evidence of Fletcher's authorship. But for the lines not deemed to be by Fletcher, Sykes felt that their depiction of women indicated that their author was not Shakespeare but Philip Massinger.

Text 2

Scholars have accepted *The Two Noble Kinsmen* as coauthored by Shakespeare since the 1970s: it appears in all major one-volume editions of Shakespeare's complete works. Though scholars disagree about who wrote what exactly, it is generally held that on the basis of style, Shakespeare wrote all of the first act and most of the last, while John Fletcher authored most of the three middle acts.

Based on the texts, both Sykes in Text 1 and the scholars in Text 2 would most likely agree with which statement?

- A. John Fletcher's writing has a unique, readily identifiable style.
- B. The women characters in John Fletcher's plays are similar to the women characters in Philip Massinger's plays.
- C. *The Two Noble Kinsmen* belongs in one-volume compilations of Shakespeare's complete plays.
- D. Philip Massinger's style in the first and last acts of *The Two Noble Kinsmen* is an homage to Shakespeare's style.

Text 1

Soy sauce, made from fermented soybeans, is noted for its umami flavor. Umami—one of the five basic tastes along with sweet, bitter, salty, and sour—was formally classified when its taste receptors were discovered in the 2000s. In 2007, to define the pure umami flavor scientists Rie Ishii and Michael O'Mahony used broths made from shiitake mushrooms and kombu seaweed, and two panels of Japanese and US judges closely agreed on a description of the taste.

Text 2

A 2022 experiment by Manon Jünger et al. led to a greater understanding of soy sauce's flavor profile. The team initially presented a mixture of compounds with low molecular weights to taste testers who found it was not as salty or bitter as real soy sauce. Further analysis of soy sauce identified proteins, including dipeptides, that enhanced umami flavor and also contributed to saltiness. The team then made a mix of 50 chemical compounds that re-created soy sauce's flavor.

Based on the texts, if Ishii and O'Mahony (Text 1) and Jünger et al. (Text 2) were aware of the findings of both experiments, they would most likely agree with which statement?

- A. On average, the diets of people in the United States tend to have fewer foods that contain certain dipeptides than the diets of people in Japan have.
- B. Chemical compounds that activate both the umami and salty taste receptors tend to have a higher molecular weight than those that only activate umami taste receptors.
- C. Fermentation introduces proteins responsible for the increase of umami flavor in soy sauce, and those proteins also increase the perception of saltiness.

- D. The broths in the 2007 experiment most likely did not have a substantial amount of the dipeptides that played a key part in the 2022 experiment.

Text 1

Africa's Sahara region—once a lush ecosystem—began to dry out about 8,000 years ago. A change in Earth's orbit that affected climate has been posited as a cause of desertification, but archaeologist David Wright also attributes the shift to Neolithic peoples. He cites their adoption of pastoralism as a factor in the region drying out: the pastoralists' livestock depleted vegetation, prompting the events that created the Sahara Desert.

Text 2

Research by Chris Brierley et al. challenges the idea that Neolithic peoples contributed to the Sahara's desertification. Using a climate-vegetation model, the team concluded that the end of the region's humid period occurred 500 years earlier than previously assumed. The timing suggests that Neolithic peoples didn't exacerbate aridity in the region but, in fact, may have helped delay environmental changes with practices (e.g., selective grazing) that preserved vegetation.

Based on the texts, how would Chris Brierley (Text 2) most likely respond to the discussion in Text 1?

- A. By pointing out that given the revised timeline for the end of the Sahara's humid period, the Neolithic peoples' mode of subsistence likely didn't cause the region's desertification
- B. By claiming that pastoralism was only one of many behaviors the Neolithic peoples took part in that may have contributed to the Sahara's changing climate
- C. By insisting that pastoralism can have both beneficial and deleterious effects on a region's vegetation and climate
- D. By asserting that more research needs to be conducted into factors that likely contributed to the desertification of the Sahara region

Text 1

Conventional wisdom long held that human social systems evolved in stages, beginning with hunter-gatherers forming small bands of members with roughly equal status. The shift to agriculture about 12,000 years ago sparked population growth that led to the emergence of groups with hierarchical structures: associations of clans first, then chiefdoms, and finally, bureaucratic states.

Text 2

In a 2021 book, anthropologist David Graeber and archaeologist David Wengrow maintain that humans have always been socially flexible, alternately forming systems based on hierarchy and collective ones with decentralized leadership. The authors point to evidence that as far back as 50,000 years ago some hunter-gatherers adjusted their social structures seasonally, at times dispersing in small groups but also assembling into communities that included esteemed individuals.

Based on the texts, how would Graeber and Wengrow (Text 2) most likely respond to the “conventional wisdom” presented in Text 1?

- A. By conceding the importance of hierarchical systems but asserting the greater significance of decentralized collective societies
- B. By disputing the idea that developments in social structures have followed a linear progression through distinct stages
- C. By acknowledging that hierarchical roles likely weren’t a part of social systems before the rise of agriculture
- D. By challenging the assumption that groupings of hunter-gatherers were among the earliest forms of social structure

Text 1

In 2007, a team led by Alice Storey analyzed a chicken bone found in El Arenal, Chile, dating it to 1321–1407 CE—over a century before Europeans invaded the region, bringing their own chickens. Storey also found that the El Arenal chicken shared a unique genetic mutation with the ancient chicken breeds of the Polynesian Islands in the Pacific. Thus, Polynesian peoples, not later Europeans, probably first introduced chickens to South America.

Text 2

An Australian research team weakened the case for a Polynesian origin for the El Arenal chicken by confirming that the mutation identified by Storey has occurred in breeds from around the world. More recently, though, a team led by Augusto Luzuriaga-Neira found that South American chicken breeds and Polynesian breeds share other genetic markers that European breeds lack. Thus, the preponderance of evidence now favors a Polynesian origin.

Based on the texts, how would the author of Text 2 most likely respond to the underlined claim in Text 1?

- A. By broadly agreeing with the claim but objecting that the timeline it presupposes conflicts with the findings of the genetic analysis conducted by Storey’s team
- B. By faulting the claim for implying that domestic animals couldn’t have been transferred from South America to the Polynesian Islands as well
- C. By critiquing the claim for being based on an assumption that before the European invasion of South America, the chickens of Europe were genetically uniform

- D. By noting that while the claim is persuasive, the findings of Luzuriaga-Neira’s team provide stronger evidence for it than the findings of the genetic analysis conducted by Storey do

Text 1

Astronomer Mark Holland and colleagues examined four white dwarfs—small, dense remnants of past stars—in order to determine the composition of exoplanets that used to orbit those stars. Studying wavelengths of light in the white dwarf atmospheres, the team reported that traces of elements such as lithium and sodium support the presence of exoplanets with continental crusts similar to Earth’s.

Text 2

Past studies of white dwarf atmospheres have concluded that certain exoplanets had continental crusts. Geologist Keith Putirka and astronomer Siyi Xu argue that those studies unduly emphasize atmospheric traces of lithium and other individual elements as signifiers of the types of rock found on Earth. The studies don’t adequately account for different minerals made up of various ratios of those elements, and the possibility of rock types not found on Earth that contain those minerals.

Based on the texts, how would Putirka and Xu (Text 2) most likely characterize the conclusion presented in Text 1?

- A. As unexpected, because it was widely believed at the time that white dwarf exoplanets lack continental crusts
- B. As premature, because researchers have only just begun trying to determine what kinds of crusts white dwarf exoplanets had
- C. As questionable, because it rests on an incomplete consideration of potential sources of the elements detected in white dwarf atmospheres

- D. As puzzling, because it's unusual to successfully detect lithium and sodium when analyzing wavelengths of light in white dwarf atmospheres

Text 1

What factors influence the abundance of species in a given ecological community? Some theorists have argued that historical diversity is a major driver of how diverse an ecological community eventually becomes: differences in community diversity across otherwise similar habitats, in this view, are strongly affected by the number of species living in those habitats at earlier times.

Text 2

In 2010, a group of researchers including biologist Carla Cáceres created artificial pools in a New York forest. They stocked some pools with a diverse mix of zooplankton species and others with a single zooplankton species and allowed the pool communities to develop naturally thereafter. Over the course of four years, Cáceres and colleagues periodically measured the species diversity of the pools, finding—contrary to their expectations—that by the end of the study there was little to no difference in the pools’ species diversity.

Based on the texts, how would Cáceres and colleagues (Text 2) most likely describe the view of the theorists presented in Text 1?

- A. It is largely correct, but it requires a minor refinement in light of the research team’s results.
- B. It is not compelling as a theory regardless of any experimental data collected by the research team.
- C. It may seem plausible, but it is not supported by the research team’s findings.
- D. It probably holds true only in conditions like those in the research team’s study.

Text 1

The fossil record suggests that mammoths went extinct around 11 thousand years (kyr) ago. In a 2021 study of environmental DNA (eDNA)—genetic material shed into the environment by organisms—in the Arctic, Yucheng Wang and colleagues found mammoth eDNA in sedimentary layers formed millennia later, around 4 kyr ago. To account for this discrepancy, Joshua H. Miller and Carl Simpson proposed that arctic temperatures could preserve a mammoth carcass on the surface, allowing it to leach DNA into the environment, for several thousand years.

Text 2

Wang and colleagues concede that eDNA contains DNA from both living organisms and carcasses, but for DNA to leach from remains over several millennia requires that the remains be perpetually on the surface. Scavengers and weathering in the Arctic, however, are likely to break down surface remains well before a thousand years have passed.

Which choice best describes how Text 1 and Text 2 relate to each other?

- A. Text 1 discusses two approaches to studying mammoth extinction without advocating for either, whereas Text 2 advocates for one approach over the other.
- B. Text 1 presents findings by Wang and colleagues and gives another research team's attempt to explain those findings, whereas Text 2 provides additional detail that calls that explanation into question.
- C. Text 1 describes Wang and colleagues' study and a critique of their methodology, whereas Text 2 offers additional details showing that methodology to be sound.
- D. Text 1 argues that new research has undermined the standard view of when mammoths went extinct, whereas Text 2 suggests a way to reconcile the standard view with that new research.

Text 1

Like the work of Ralph Ellison before her, Toni Morrison's novels feature scenes in which characters deliver sermons of such length and verbal dexterity that for a time, the text exchanges the formal parameters of fiction for those of oral literature. Given the many other echoes of Ellison in Morrison's novels, both in structure and prose style, these scenes suggest Ellison's direct influence on Morrison.

Text 2

In their destabilizing effect on literary form, the sermons in Morrison's works recall those in Ellison's. Yet literature by Black Americans abounds in moments where interpolated speech erodes the division between oral and written forms that literature in English has traditionally observed. Morrison's use of the sermon is attributable not only to the influence of Ellison but also to a community-wide strategy of resistance to externally imposed literary conventions.

Based on the texts, how would the author of Text 2 most likely characterize the underlined claim in Text 1?

- A. As failing to consider Ellison's and Morrison's equivalent uses of the sermon within the wider cultural context in which they wrote
- B. As misunderstanding the function of sermons in novels by Black American writers other than Ellison and Morrison
- C. As disregarding points of structural and stylistic divergence between the works of Ellison and those of Morrison
- D. As being indebted to the tradition of resisting literary conventions that privilege written forms, such as novels, over sermons and other oral forms